

Approximate Mean Response Time for Heterogeneous Two-Queue Fork-Join Systems

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Abstract

We derive a simple closed-form approximation for the mean response time of a heterogeneous two-queue M/M/1 fork-join system, where servers may have different service rates. The approximation interpolates between an independent upper bound and a bottleneck lower bound using a weight determined by the average server utilization. The formula is exact for the homogeneous case (recovering the Nelson–Tantawi result), satisfies known bounds, and exhibits correct limiting behavior in both light and heavy traffic. Discrete-event simulation validates the approximation to within 2% across a wide range of parameters.

1 Introduction

Consider a fork-join (FJ) queueing system with two parallel servers having service rates μ_1 and μ_2 . Jobs arrive as a Poisson process with rate λ . Each arriving job forks into two tasks, one dispatched to each server; the job completes (joins) only when *both* tasks have finished service. Each server operates as an independent M/M/1 queue with FCFS discipline.

The primary performance measure is the mean job response time:

$$T = \mathbb{E}[\max(R_1, R_2)], \quad (1)$$

where R_i is the sojourn time of the task at server i . The system is stable when $\lambda < \min(\mu_1, \mu_2)$.

For the *homogeneous* case ($\mu_1 = \mu_2 = \mu$), Nelson and Tantawi [6] derived the exact result:

$$T_2^{\text{hom}} = \frac{12 - \rho}{8(\mu - \lambda)} = \frac{12 - \rho}{8} \cdot T_1, \quad (2)$$

where $\rho = \lambda/\mu$ and $T_1 = 1/(\mu - \lambda)$ is the M/M/1 mean sojourn time.

For *heterogeneous* servers ($\mu_1 \neq \mu_2$), no closed-form expression for T is known. Flatto and Hahn [3, 2] obtained the exact generating function via elliptic function parametrization, but extracting moments requires solving boundary value problems. Various bounds have been established [1, 5, 4, 7], but none yield a simple evaluable expression.

In this paper, we propose a closed-form approximation that:

1. is exact for the homogeneous case,
2. satisfies known upper and lower bounds,
3. has correct light-traffic and heavy-traffic limits, and
4. agrees with simulation to within 2%.

2 Bounds

We use the notation $\rho_i = \lambda/\mu_i$ for the utilization of server i and $T_i = 1/(\mu_i - \lambda)$ for the mean M/M/1 sojourn time at server i .

2.1 Independent Upper Bound

If the sojourn times R_1 and R_2 were independent—each exponentially distributed with rate $\mu_i - \lambda$ —then:

$$T_{\text{UB}} = \frac{1}{\mu_1 - \lambda} + \frac{1}{\mu_2 - \lambda} - \frac{1}{\mu_1 + \mu_2 - 2\lambda}. \quad (3)$$

In reality, R_1 and R_2 are positively correlated (shared Poisson arrivals), which reduces $\mathbb{E}[\max(R_1, R_2)]$ below the independent case. Hence $T \leq T_{\text{UB}}$ [1].

2.2 Bottleneck Lower Bound

By Jensen's inequality (max is convex):

$$T_{\text{bot}} = \max(T_1, T_2) = \max\left(\frac{1}{\mu_1 - \lambda}, \frac{1}{\mu_2 - \lambda}\right) \leq T. \quad (4)$$

2.3 Split-Merge Upper Bound

A split-merge system forces servers to synchronize after each job, making it an upper bound on fork-join [8]. Treating $\max(S_1, S_2)$ as the effective service time of an M/G/1 queue, the Pollaczek–Khinchine formula gives:

$$T_{\text{SM}} = \mathbb{E}[S_{\text{max}}] + \frac{\lambda \mathbb{E}[S_{\text{max}}^2]}{2(1 - \lambda \mathbb{E}[S_{\text{max}}])}, \quad (5)$$

where for independent $S_i \sim \text{Exp}(\mu_i)$:

$$\mathbb{E}[S_{\text{max}}] = \frac{1}{\mu_1} + \frac{1}{\mu_2} - \frac{1}{\mu_1 + \mu_2}, \quad (6)$$

$$\mathbb{E}[S_{\text{max}}^2] = \frac{2}{\mu_1^2} + \frac{2}{\mu_2^2} - \frac{2}{(\mu_1 + \mu_2)^2}. \quad (7)$$

This bound requires $\lambda \mathbb{E}[S_{\text{max}}] < 1$, which is more restrictive than the FJ stability condition.

2.4 Summary of Bounds

For all stable parameters:

$$T_{\text{bot}} \leq T \leq T_{\text{UB}}. \quad (8)$$

The split-merge bound $T \leq T_{\text{SM}}$ also holds when $\lambda \mathbb{E}[S_{\text{max}}] < 1$.

3 Proposed Approximation

3.1 Derivation Strategy

We seek an interpolation between T_{UB} and T_{bot} that satisfies:

1. **Exactness:** Reduces to (2) when $\mu_1 = \mu_2$.

2. **Light traffic** ($\lambda \rightarrow 0$): $T \rightarrow T_{\text{UB}}$ (nearly independent tasks).
3. **Heavy traffic** ($\rho \rightarrow 1$): $T \rightarrow T_{\text{bot}}$ (bottleneck dominance).
4. **Bound compliance**: $T_{\text{bot}} \leq T \leq T_{\text{UB}}$ for all valid parameters.

3.2 The Formula

Define the average utilization $\bar{\rho} = (\rho_1 + \rho_2)/2$ and the interpolation weight:

$$\alpha = \frac{\bar{\rho}}{4} = \frac{\rho_1 + \rho_2}{8}. \quad (9)$$

Then:

$$\boxed{T \approx (1 - \alpha) T_{\text{UB}} + \alpha T_{\text{bot}}} \quad (10)$$

where T_{UB} and T_{bot} are given by (3) and (4).

Equivalently:

$$T \approx \left(1 - \frac{\rho_1 + \rho_2}{8}\right) T_{\text{UB}} + \frac{\rho_1 + \rho_2}{8} T_{\text{bot}}. \quad (11)$$

3.3 Verification: Homogeneous Case

When $\mu_1 = \mu_2 = \mu$, we have $\rho_1 = \rho_2 = \rho$, $\alpha = \rho/4$, and:

- $T_{\text{UB}} = 2T_1 - T_1/2 = \frac{3}{2} T_1$
- $T_{\text{bot}} = T_1$

Substituting into (10):

$$T = \left(1 - \frac{\rho}{4}\right) \frac{3}{2} T_1 + \frac{\rho}{4} T_1 = \frac{3}{2} T_1 - \frac{3\rho}{8} T_1 + \frac{\rho}{4} T_1 = \frac{12 - \rho}{8} T_1, \quad (12)$$

which exactly recovers the Nelson–Tantawi result (2).

3.4 Properties

Table 1: Properties of the approximation.

Property	Status
Exact for homogeneous case ($\mu_1 = \mu_2$)	✓
Satisfies $T_{\text{bot}} \leq T \leq T_{\text{UB}}$	✓ ($0 \leq \alpha \leq 1/4 < 1$)
Correct light traffic limit ($\lambda \rightarrow 0$)	✓ ($\alpha \rightarrow 0$, $T \rightarrow T_{\text{UB}}$)
Correct heavy traffic limit ($\rho \rightarrow 1$)	✓ (T_{bot} diverges; $T/T_{\text{bot}} \rightarrow 1$)
Closed-form, directly evaluable	✓

Table 2: Approximation vs. simulation for heterogeneous fork-join ($n = 2$).

μ_1	μ_2	λ	ρ_1	T_{bot}	T_{sim}	T_{approx}	T_{UB}	Error
1.0	1.0	0.3	0.30	1.429	2.090	2.089	2.143	−0.05%
1.0	1.0	0.6	0.60	2.500	3.564	3.563	3.750	−0.04%
1.0	1.0	0.9	0.90	10.00	13.82	13.88	15.00	+0.42%
1.0	1.5	0.3	0.30	1.429	1.707	1.716	1.736	+0.55%
1.0	1.5	0.6	0.60	2.500	2.765	2.799	2.842	+1.22%
1.0	1.5	0.9	0.90	10.00	10.01	10.19	10.24	+1.82%
1.0	2.0	0.3	0.30	1.429	1.584	1.591	1.600	+0.42%
1.0	2.0	0.6	0.60	2.500	2.621	2.641	2.659	+0.75%
1.0	2.0	0.9	0.90	10.00	9.918	10.06	10.08	+1.46%
1.0	3.0	0.3	0.30	1.429	1.498	1.501	1.505	+0.18%
1.0	3.0	0.6	0.60	2.500	2.545	2.554	2.560	+0.34%
1.0	3.0	0.9	0.90	10.00	9.886	10.02	10.02	+1.34%
1.0	5.0	0.3	0.30	1.429	1.455	1.455	1.456	+0.03%
1.0	5.0	0.6	0.60	2.500	2.513	2.517	2.519	+0.17%
1.0	5.0	0.9	0.90	10.00	9.876	10.01	10.01	+1.31%

4 Simulation Validation

The approximation was validated against discrete-event simulation of the two-queue FJ system (2M jobs per scenario, 100K warmup). Results across a range of heterogeneity levels (μ_2/μ_1 from 1 to 5) and loads (ρ_1 from 0.3 to 0.9) are shown in Table 2.

Key observations:

- The approximation is **exact** for the homogeneous case (errors $< 0.05\%$ at low/moderate load, $< 0.5\%$ at high load due to simulation noise).
- For heterogeneous cases, the error is consistently **positive** (slight overestimation), staying within $+2\%$ across all tested scenarios.
- The largest error occurs at moderate heterogeneity ($\mu_2/\mu_1 \approx 1.5$) under heavy load ($\rho_1 = 0.9$), reaching $+1.8\%$.
- For strong heterogeneity ($\mu_2/\mu_1 \geq 3$), errors are small because T_{UB} and T_{bot} converge (the bottleneck dominates).

Figure 1 shows the approximation, bounds, and simulation results as a function of load for a heterogeneous system with $\mu_1 = 1.0$, $\mu_2 = 2.0$. Figure 2 shows the same quantities as a function of the heterogeneity ratio μ_2/μ_1 at fixed load $\lambda = 0.6$.

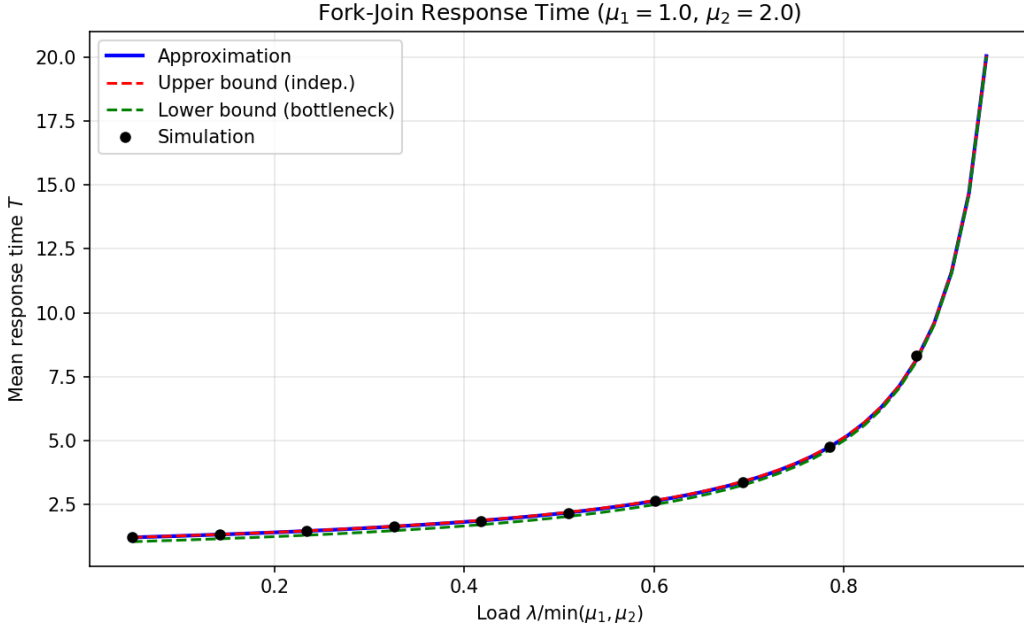


Figure 1: Mean response time vs. load ($\lambda/\min(\mu_1, \mu_2)$) for $\mu_1 = 1.0$, $\mu_2 = 2.0$.

5 Discussion

5.1 Intuition

The interpolation weight $\alpha = \bar{\rho}/4$ captures the effect of queueing correlation. At low load, tasks rarely queue, so their sojourn times are nearly independent and the upper bound is tight. As load increases, tasks experience correlated waiting due to the shared arrival stream, which reduces the synchronization overhead and pulls the response time toward the bottleneck.

The factor of $1/4$ in α is inherited from the homogeneous case analysis. In the homogeneous setting, the synchronization delay is $\text{Sync} = (4 - \rho)/[8(\mu - \lambda)]$ [6], and the independent synchronization delay would be $1/[2(\mu - \lambda)]$. Their ratio at $\rho = 1$ determines the interpolation scaling.

5.2 Comparison with Other Approaches

Table 3 summarizes existing approaches for fork-join response time analysis.

Our approximation fills a gap: it is the only directly evaluable closed-form expression for heterogeneous fork-join (not split-merge) that is exact in the homogeneous limit and respects known bounds.

5.3 Limitations

- The formula is a heuristic approximation, not an exact result. The weight $\alpha = \bar{\rho}/4$ is a natural generalization from the homogeneous case but is not derived from first principles for

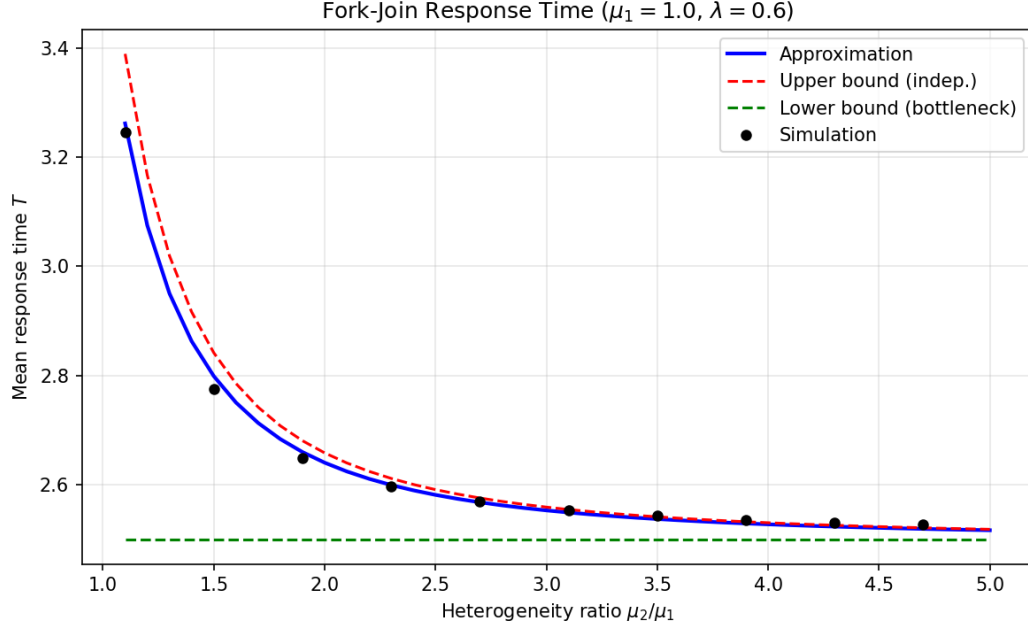


Figure 2: Mean response time vs. heterogeneity ratio (μ_2/μ_1) for $\mu_1 = 1.0$, $\lambda = 0.6$.

Table 3: Comparison of approaches for fork-join mean response time.

Approach	Type	Closed-form?	Heterogeneous?
Nelson–Tantawi [6]	Approximation	Yes	No
Flatto–Hahn [3, 2]	Exact (gen. func.)	No	Yes
Ko–Serfozo [5]	Bounds	Semi	Yes
Kemper–Mandjes [4]	Bounds + heavy traffic	Semi	Yes
Rizk et al. [7]	Distributional bounds	Semi	Yes
Varki [8]	Split-merge (M/G/1)	Yes	Yes (UB for FJ)
This work	Interpolation approx.	Yes	Yes

the heterogeneous setting.

- The approximation consistently overestimates the true response time by up to $\sim 2\%$, with the largest errors at moderate heterogeneity ($\mu_2/\mu_1 \approx 1.5$) under heavy load. This suggests the correlation correction could be slightly stronger in the heterogeneous case.
- Further validation against numerical inversion of the Flatto–Hahn generating function could provide exact reference values for calibration.

6 Conclusion

For a heterogeneous two-queue M/M/1 fork-join system with service rates μ_1 , μ_2 and Poisson arrival rate λ , the mean response time is well approximated by:

$$T \approx \left(1 - \frac{\rho_1 + \rho_2}{8}\right) T_{\text{UB}} + \frac{\rho_1 + \rho_2}{8} T_{\text{bot}}, \quad (13)$$

where T_{UB} and T_{bot} are defined in (3) and (4). This expression is exact for homogeneous servers, satisfies known bounds, and agrees with simulation to within 2% across a wide range of parameters.

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